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- Problem Statement ID – 25045
- Problem Statement Title- Blockchain-Based Supply Chain Transparency for Agricultural Produce
- Theme- Agriculture, FoodTech & Rural Development
- PS Category- Software
- Team ID - 97277
- Team Name - PARASTECH 6

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Blockchain-based Agricultural Supply Chain Transparency System

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★ Proposed Solution:

- Develop a **Decentralized**, end-to-end **supply chain** system leveraging **blockchain** technology to meticulously track agricultural produce.
- The system will track products from their origin on the farm all the way to the final consumer.
- This solution will guarantee the authenticity, **Transparency**, and fair pricing of all goods within the supply chain.
- It will provide farm-to-fork **Traceability** using a combination of blockchain and unique **QR codes** assigned to each product.

Innovation & Uniqueness:

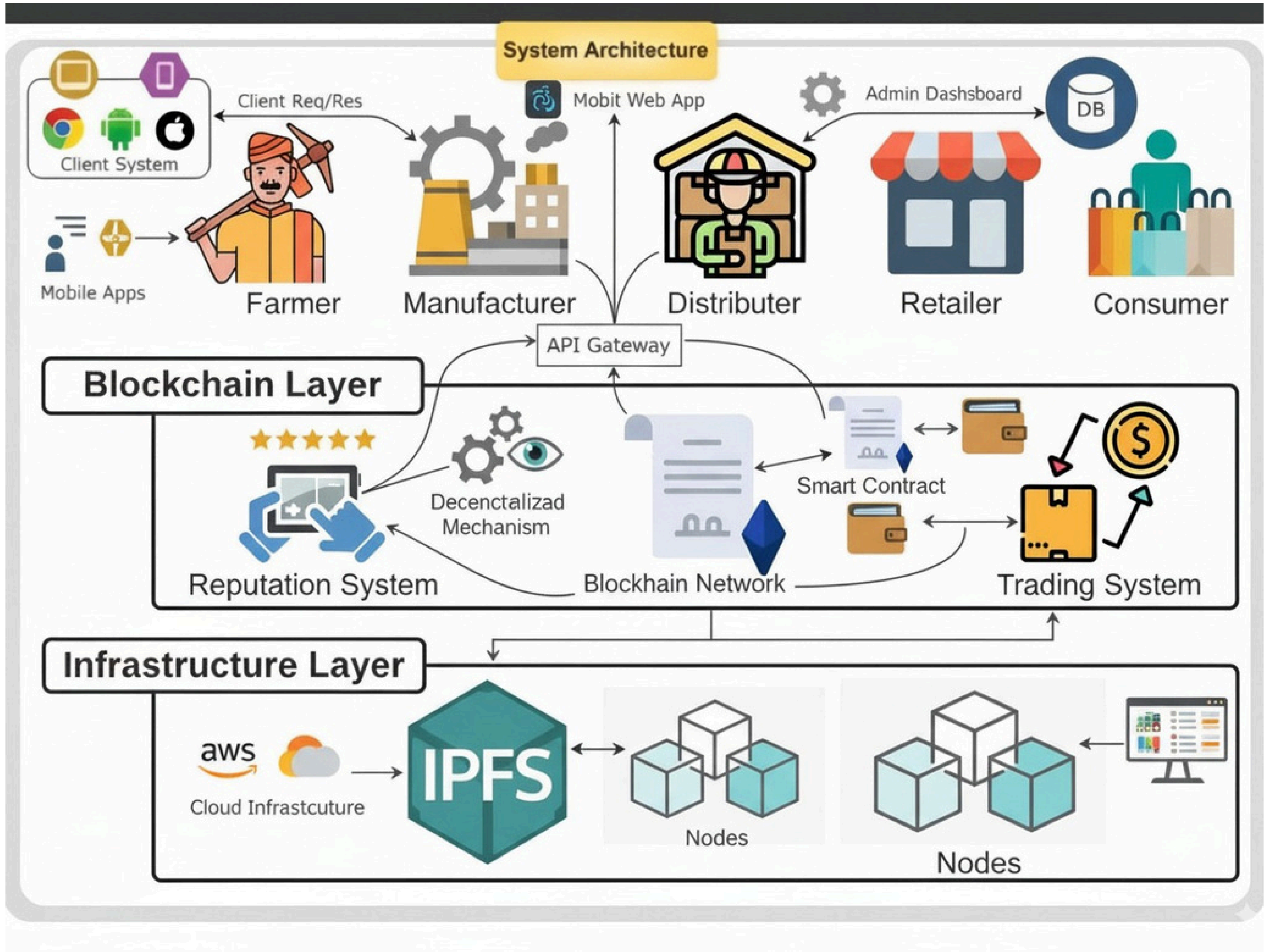
- Immutable Record Keeping: The use of blockchain ensures that all records are tamper-proof, creating a permanent and verifiable history for every product. This eliminates fraud and builds **Trust.**
- Consumer Verification via QR Codes: Consumers can scan a **QR code** on the product packaging using a dedicated mobile app to instantly access its complete history, including the farm of origin, certifications, and
- transportation details.

Automated Processes with Smart Contracts:

- **Smart contracts** will be used to automate key transactions.
- These include **Automated Payments** from distributors to farmers upon delivery.
- They will also manage automated transfers of ownership and the issuance of digital certifications (e.g., organic, fair trade**).**

Decentralized Data Storage:

- **IPFS** (InterPlanetary File System) will be integrated to store large, detailed files.
- This includes high-resolution images of the farm, certification documents, and quality control reports without burdening the blockchain.
- Real-time **Consumer Engagement**: A user-friendly mobile application will be developed to provide consumers with real-time traceability. They can follow their food's journey, from the moment it's harvested to its arrival on the store shelf, fostering a deeper connection to their food source.



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TECHNICAL APPROACH

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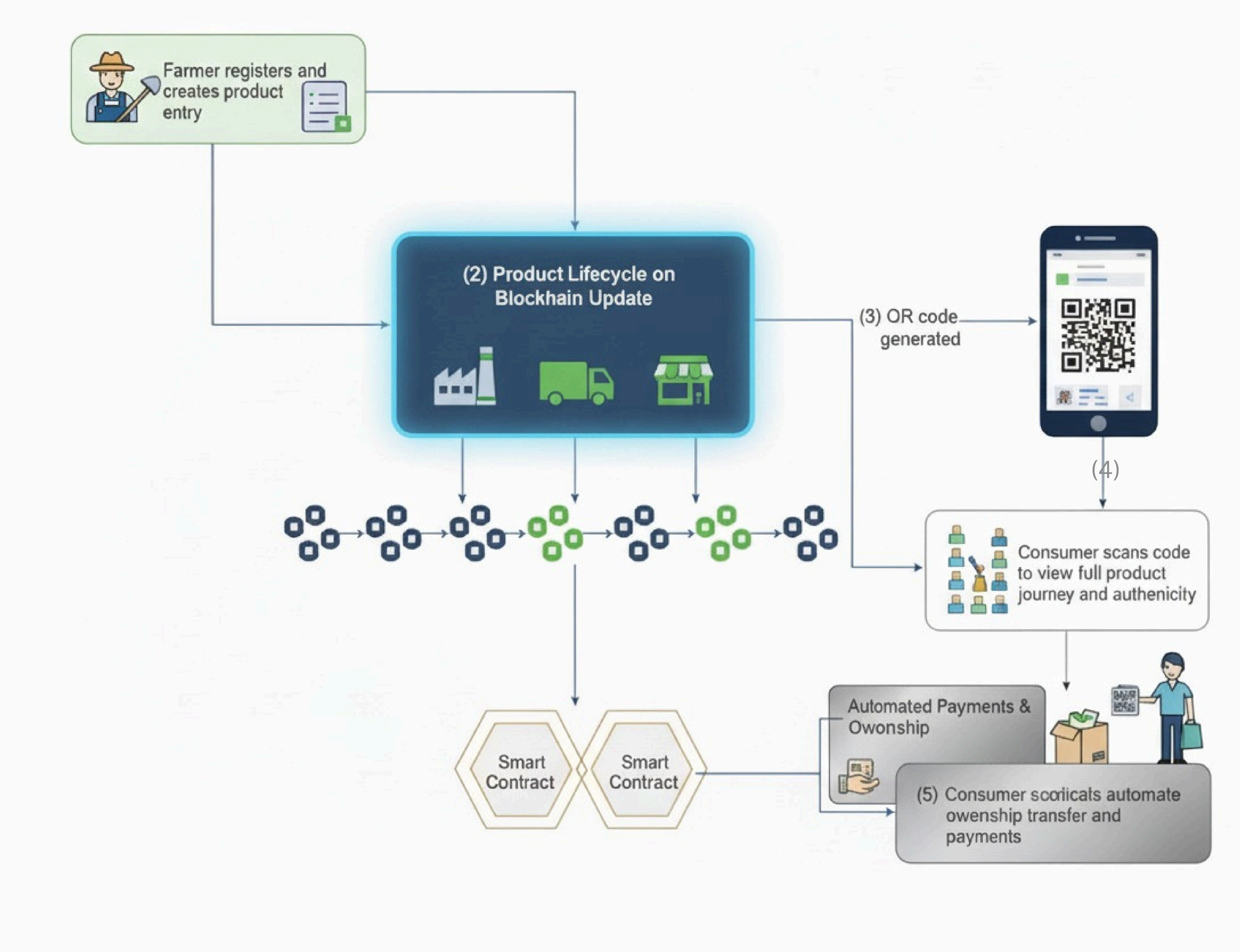
★TECHNOLOGY STACK

- **Blockchain Platforms –** Ethereum (Testnet), Polygon, Hyperledger
- **Smart Contracts & Frameworks –** Solidity, Chaincode, Hardhat
- **Frontend & UI –** React.js, Web3.js, Tailwind CSS
- **Backend & APIs –** Node.js, Express.js, GraphQL
- **Storage & Databases –** IPFS, MongoDB, Firebase
- **Deployment & Hosting –** AWS, Netlify, Docker



Docker · TRUFFLE · Firebase · GitHub · JS · Solidity · METAMASK

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FEASIBILITY AND VIABILITY

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Feasiblity:

1. **Ethereum Testnet**: Enables rapid, low-cost **prototyping.**
2. Free Hosting: Platforms like Vercel/Netlify simplify deployment.
3. Enterprise Scalability: **Polygon & Hyperledger Fabric** handle large-scale operations.
4. Mobile Accessibility: Lightweight apps make the platform usable for farmers.

Viability:

1. Transparent Supply Chain: Farmers can track produce from farm to market.
2. Fair Pricing: **Smart contracts** ensure automatic payments and agreements
3. Revenue Potential: Transaction fees, subscriptions, and premium services
4. Market Expansion: **Scalable** to multiple regions, crops, and stakeholders.

Novelty:

1.First-of-its-kind blockchain platform ensuring **authenticity** and **fairness** from farm to consumer.

2.Combines **traceability**, fair pricing, and **data-driven insights** in a single system.

3.Uses smart contracts to automate payments and agreements.

4.Real-time monitoring of produce via **loT integration**

Cost Reduction:

1.**Eliminates middlemen**, ensuring higher farmer profits.

2.Layer-2 blockchain reduces transaction costs while ensuring trust.

3.Automation reduces paperwork and admin overhead.

4.Transparent tracking **minimizes spoilage** and **fraud losses.**

Scalability:

1.Easily expands across regions, crops, and stakeholders.

2.Integrates with **government bodies, NGOs**, and **agri-businesses.**

3.API-friendly for third-party features and analytics.

4.Handles growing transactions without performance issues.

Supporting Facts for Feasibility and Viability

1. Food Fraud Costs: Food fraud causes estimated $40 billion annual losses worldwide, highlighting the urgent need for transparent supply chains.

2. Consumer Demand: 73% of global consumers say they want to know more about the origin and journey of their food products.

3. Farmer Exploitation: Middlemen often capture 70–80% of the final product value, leaving farmers with unfair profits.

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IMPACT AND BENEFITS

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★ Benefits of the solution

🤝 Social Impact

1. Prevents farmer exploitation by **eliminating middlemen.**
2. Increases consumer trust with **farm-to-fork transparency.**
3. Promotes fair trade and community empowerment.

💰 Economic Benefits

1. Reduces fraud (food fraud costs ~$40B annually).

2. Ensures faster and automated payments through smart contracts.

3. Opens new market **opportunities for authentic** and **certified produce.**

🌱 Environmental Impact

1..Tracks **sustainability** and organic certifications.

2. Measures **carbon footprint** across the supply chain.

3. Encourages eco-friendly farming and distribution practices.

📚 Educational Advantage

1. Increases farmer awareness of digital and **blockchain tools.**
2. Educates consumers about food sourcing and safety.
3. Provides training opportunities in emerging **AgriTech fields.**

⚙️ Technological / Operational Advantage

1. Improves traceability with blockchain and QR codes.
2. Reduces paperwork with smart contracts and **IPFS storage**.
3. Scales easily from local farmer groups to enterprise-level supply chains.

✦Potential impact on the target audience✦

1.Farmer Income Growth: Direct sales and transparent pricing can increasefarmer income by 20–30%.

2. Fraud Reduction: Blockchain traceability can reduce food fraud and middlemen exploitation by 40–50%.

3. Consumer Trust: QR-based product verification can boost consumer confidence and authentic purchases by 30–40%..

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RESEARCH AND REFERENCES

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Reference Platforms :

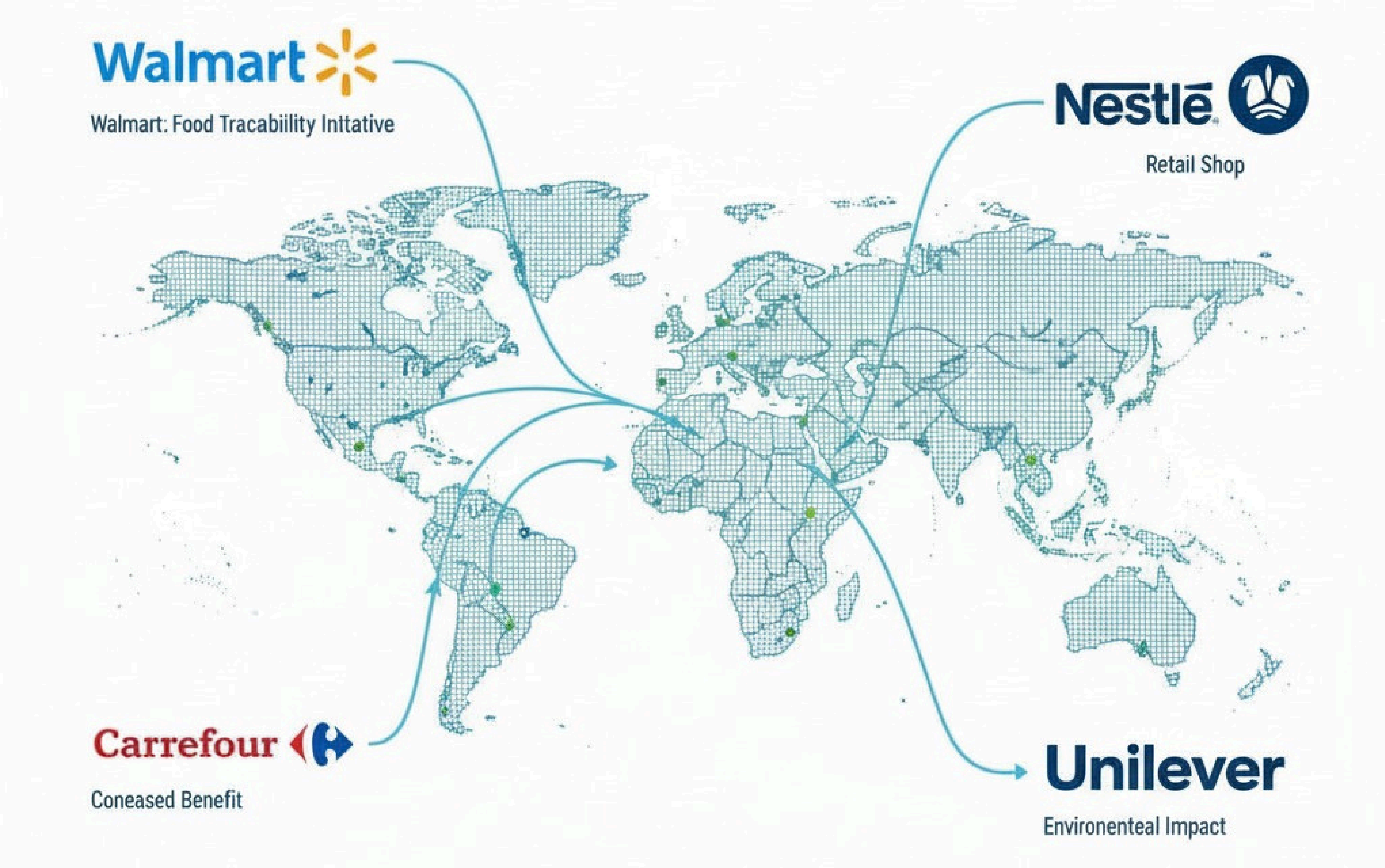
- AGRI DIGITAL
- TE FOOD
- SUPPLY CHAIN DIGITAL

RESEARCH AND PRACTICES :

- Blockchain-Based Traceability (MDPI)
- QR code in supply chains
- Ethereum/blockchain implementation for agriculture

FEASIBILITY FACTS :

- LINKEDIN SURVEY
- IBM Food Trust Case Study
- World Economic Forum Report



GLOBAL BLOCKCHAIN ADOPTION IN AGRICULTURE

YOUTUBE DEMO : https://youtu.be/CfKZAM7riL4